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**Health information systems and the information
cycle: What is important for data scientists?**

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Dar es Salaam

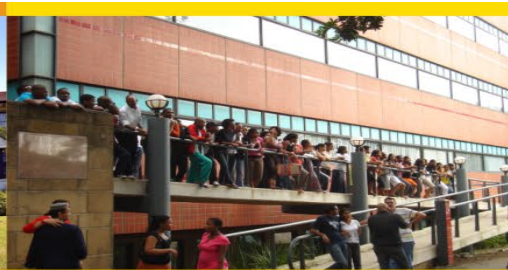
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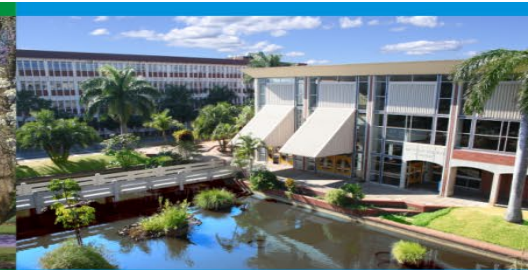
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Learning objectives

By the end, students should be able to

- Map a DHIS data element through its full information pathway.
- Identify transformation points where data can be lost, duplicated, reclassified or strategically reported.
- Distinguish source data, aggregated data, indicators and decisions.
- Diagnose a surprising indicator by tracing it back to the source.

What is a District Health Information System (DHIS)?

A theoretical starting point for data scientists

- A Health Information System (HIS) is a system that generates, manages, analyses and communicates information about individual and population health.
- A District Health Information System (DHIS) operationalises this function at district level by bringing together routine service information from health facilities and programmes.
- Its purpose is not simply to store data: information should support planning, management, monitoring, evaluation and improvement of health services.
- **Key distinction:** the DHIS is both a technical information system and a social/organisational system involving people, definitions, routines and decisions.

DEFINITION

- That system which transmits information about the health of individuals/ populations



Figure 1 The six components of a Health Information System

DHIS as part of the health system

Information is an enabling function, not a stand-alone activity

HEALTH SERVICE DELIVERY

Records encounters, services, diagnoses, prevention and programme activities.

HEALTH WORKFORCE

Health workers generate, interpret and use information in routine work.

MANAGEMENT

Managers use information to assess performance, allocate resources and respond to problems.

FINANCING / RESOURCES

Information supports planning for staffing, commodities, infrastructure and budgets.

GOVERNANCE

Information supports accountability, monitoring, coordination and stewardship.

The DHIS information cycle

From events in the real world to decisions — and back again



- The original presentation describes a cyclical process of information generation, management, dissemination, evaluation and review.
- The cycle means that information is not the endpoint: its value is realised when it changes understanding, decisions or action.
- For data scientists, every transition is a potential transformation of the underlying data.

How does a DHIS strengthen the health system?

Five linked functions

1. VISIBILITY

Makes service activity and population needs more visible to managers.

2. ACCOUNTABILITY

Allows performance to be monitored against indicators and targets.

3. PLANNING

Supports assessment of needs and allocation of limited resources.

4. LEARNING

Creates feedback loops through analysis, review and repeated measurement.

5. COORDINATION

Links facility, district, provincial and national information needs.

- The important theoretical point: information strengthens the health system when it is connected to management processes and action.

A DHIS is not a mirror of reality

It is a measurement system

- The health system produces events: patients attend, services are delivered, diagnoses are made and commodities are used.
- The DHIS selects which events become data, defines how they are counted, aggregates them and converts them into indicators.
- Therefore, every DHIS number reflects choices about definitions, inclusion, recording, aggregation, denominators and reporting.
- **This creates the bridge to today's data-science perspective:** ask not only “What does the indicator show?” but “What processes produced the indicator?”

TRANSITION TO THE INTERACTIVE PART

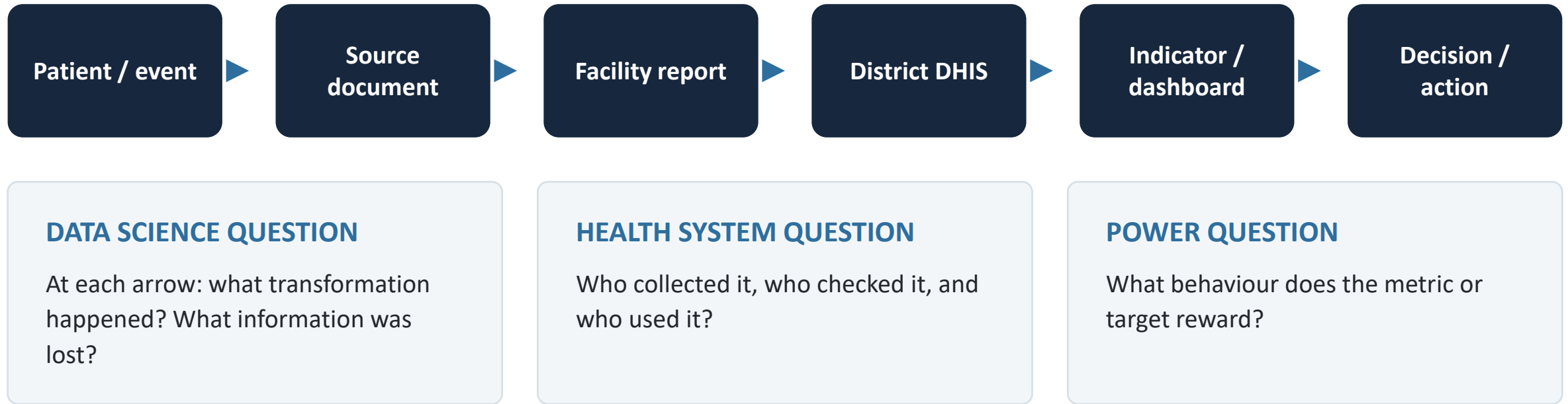
We will now follow one data element from → facility → district → national use and identify where the value can change?

DHIS as a Data-Generating System

- data → facility → district → province/national use
- Where definitions, aggregation, denominators and incentives can change the data

The central idea

The DHIS is not just a database — it is a chain of transformations



The original information cycle



- **Interactive prompt:** Which step would you audit first if a national indicator suddenly changed?
- Hint: do not assume the error is at the point where you first see it.

What counts as 'source data'?

PATIENT RECORD

Identity + clinical details,
diagnosis, procedures,
treatment and tests.

TALLY SHEET

Counts of identical
events/services where
follow-up is not required.

REGISTER

Continuity data for ANC,
immunisation, FP, TB,
chronic care, etc.

REPORT

Selected data elements
sent onward to the
district.

- **DATA SCIENTIST CHECK:** Are these four sources measuring the same unit of observation?
- If not, aggregation can create apparently precise but conceptually incompatible totals.

Interactive 1 — Follow one patient

Work in pairs • 4 minutes

- Scenario: A patient attends a PHC facility for hypertension care.
- Trace: consultation → patient-held record → chronic register → monthly facility report → district dataset → indicator.
- Mark every point where a person or system can change the value.
- **QUESTION:** What is the smallest unit of data you would want to retain for auditability?

Where can the value change?

Think of each transition as a data transformation

1 • DEFINITION

What exactly qualifies as a case, visit, completed service or new patient?

2 • RECORDING

Was the event recorded once, twice or not at all?

3 • COLLATION

Are subtotals added correctly without double-counting?

4 • REPORTING

Which fields are selected for the monthly report?

5 • AGGREGATION

What happens when facility data become district totals?

6 • DENOMINATOR

Which population is used to turn counts into rates?

7 • INCENTIVES

What gets rewarded, monitored or penalised?

Definitions are part of the data-generating process

A number is only meaningful if the counting rule is stable

- Example: “hypertension patient seen” could mean a registered patient, a consultation, a follow-up visit, or a unique patient in a month.
- A change in definition can create a trend break even when underlying service use is unchanged.
- Important to have common definitions and effective indicators.
- **DATA SCIENCE CHECK:** version the definition alongside the value.

Interactive 2 — Definition trap

Vote before discussing

FACILITY A

100 hypertension consultations; 70 unique patients

FACILITY B

100 hypertension consultations; 95 unique patients

- If the indicator is “patients seen”, can the two facilities report 100 each?
- If the indicator is “consultations”, should they report 100 each?
- What metadata would you need to interpret the numbers?

Aggregation is not neutral

explicit warnings about duplication



- Example from the source: a headcount recorded at the entrance must not also be added to headcounts from each examining room.
- **DATA SCIENCE CHECK:** Every aggregation rule should state the unit being summed and the expected grain.

Interactive 3 — Find the double count

Facility reports 600 visits

ENTRANCE HEADCOUNT

600 clients entered the facility

CURATIVE ROOMS

420 consultations

MATERNITY + OTHER

180 service contacts

- **Question:** Is $600 + 420 + 180 = 1\,200$ a legitimate facility total?
- What information would you ask for before aggregating?
- Which variable identifies the unit of count?

Denominators turn counts into indicators

What is an indicator? converting raw data into comparable information

NUMERATOR

Number receiving a service / experiencing an event

DENOMINATOR

Target population or reference population

- Indicator = numerator ÷ denominator (often × 100 or × 1 000).
- A correct numerator with the wrong denominator can produce a misleading indicator.
- **DATA SCIENCE CHECK:** Store numerator, denominator, definition, reference period and geography — not only the final percentage.

INDICATORS

- Convert raw data into useful information;
- Are observable markers of progress towards defined targets;
- Are used to describe the situation and to measure changes over time;
- Provide information about a broad range of conditions through a single measure;
- Provide a yardstick whereby institutions or teams can compare themselves to others doing similar work.

Indicators enable us to compare “apples” with “apples”, not with “melons”!

CLASSIFICATION OF INDICATORS

There are four main types of indicators (WHO 2000):

Type of indicator	Description	Example
Count Indicator	Number of events without denominator	Number of new cases of Acute Flaccid Paralysis
Proportion Indicator	Numerator is contained in the denominator	Proportion of health centres without electricity
Rate Indicator	Frequency of the event in a specified time in a given population	Incidence of new TB cases in a given population per year
Ratio Indicator	Numerator is not included in the denominator	Ratio of nurses to population Ratio of male TB deaths to female TB deaths

Table 9: Indicator Type

IDEAL INDICATOR

Reliable

Gives the same results if used by different people in different places

Appropriate

Fits in with local needs, capacity and culture and the decisions to be made

Valid

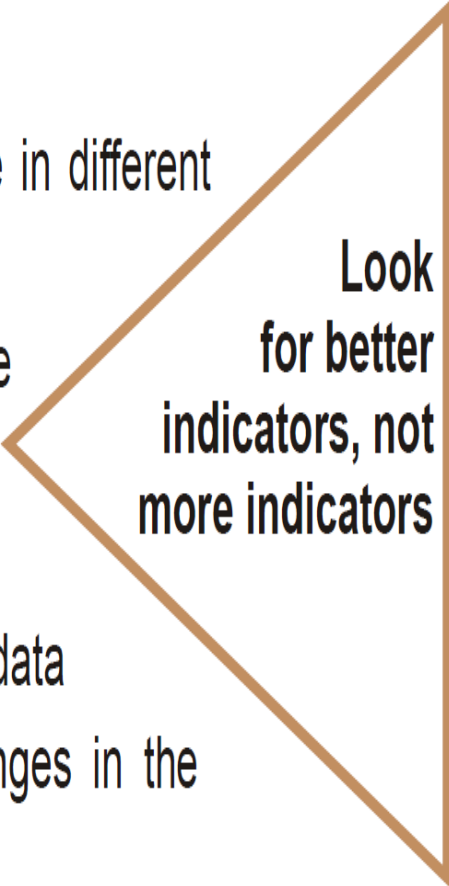
Truly measures what is of interest

Easy

Able to be simply calculated using routinely available data

Sensitive

Changes in the indicator immediately reflect changes in the actual situation under study



**Look
for better
indicators, not
more indicators**

The target population is a hidden data model

Who should be counted matters as much as who was counted

- Each priority programme has a different target group, defined by age, sex and risk factors.
- A facility report tells you about users; it does not automatically tell you about non-users.
- Coverage therefore depends on both service data and a defensible estimate of the population that should have received the service.
- **Interactive question:** What happens to coverage if the denominator is underestimated by 20%?

Interactive 4 — Same numerator, different story

FACILITY X

800 children immunised
Denominator = 1 000
Coverage = 80%

REVISED POPULATION

800 children immunised
Denominator = 1 250
Coverage = 64%

QUESTION

Did service delivery change?
Or did the denominator change?

- Discuss: Which version should a data scientist publish? What evidence is needed first?

Facility → district → national

Different levels use information for different purposes

Facility

manage services

District

compare / plan

Province

allocate / monitor

National

policy / stewardship

- **“Information pyramid”**: each level should receive data relevant to its work while avoiding excessive data that cannot be acted upon.
- **Key implication for data scientists**: aggregation changes the analytical questions that can be answered.

What gets lost when we aggregate?

A district total can be accurate and still hide important structure

TIME

Daily variation → monthly total

PLACE

Facility variation → district total

PEOPLE

Patient trajectories → counts

CONTEXT

Reason / notes → selected fields

- **Interactive prompt:** Which of these losses could create an ecological fallacy or conceal an access problem?

Incentives can alter what gets recorded

Data quality is partly a behavioural phenomenon

- Targets and norms can make particular indicators highly visible to managers.
- If performance is judged on a metric, staff may have reasons to prioritise, classify or report activity in ways that improve the metric.
- This does not imply intentional fraud: workload, uncertainty, definitions, reporting routines and feedback can all shift recorded values.
- **DATA SCIENCE CHECK:** Treat the metric as an output of both service activity AND the measurement system.

Interactive 5 — The target changes the data

Small-group exercise • 5 minutes

- A programme target is attached to the number of patients receiving a service.
- You observe a sudden rise immediately before reporting deadlines.
- Generate 4 competing explanations:
 - 1. True increase in service use
 - 2. Improved recording / completeness
 - 3. Reclassification or changed definition
 - 4. Behaviour shaped by the target or reporting process
- What additional data would discriminate between these explanations?

Data quality: not just ‘clean vs dirty’

correctness, completeness and consistency

CORRECTNESS

Are values plausible and correctly recorded?

COMPLETENESS

Did all units/facilities submit all required data?

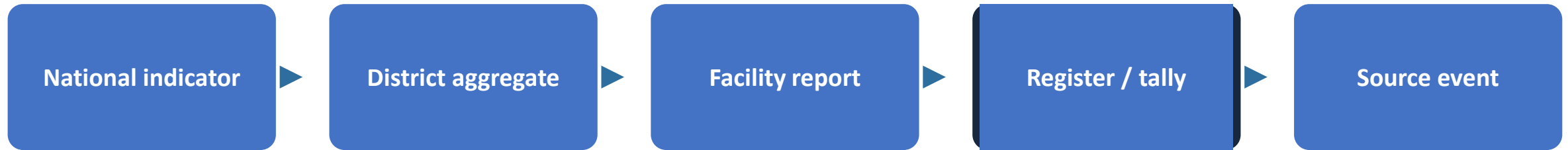
CONSISTENCY

Are definitions/ranges comparable across facilities?

- Data audit: return to the register, tally sheet or patient-held record when a value is questioned.
- For data scientists, preserve the original value and the corrected value where possible; keep an audit trail.

A practical audit trail

Every published indicator should be traceable backwards



- Reverse-tracing question: “Show me the records that contributed to this number.”
- Forward-tracing question: “Show me where this source event appears in the final indicator.”

Interactive 6 — Trace the anomaly

Case: district hypertension visits ↑ 35% in one month

- Your task: design a diagnostic tree before looking at the answer.
- Check in order: definition → completeness → duplicates → facility mix → aggregation → denominator → reporting deadline → true service change.
- Which check is most informative? Which is cheapest? Which is most likely to reveal a system-level problem?
- Share your diagnostic tree with another group.

Indicators are models

An indicator compresses many assumptions into one number

RAW EVENT

Patient / service / case

COUNTING RULE

Definition + inclusion

AGGREGATION

Who/where/when

INDICATOR

Numerator / denominator

- The final number is therefore not “the data”; it is the result of a sequence of design and operational choices.

The data scientist's DHIS checklist

- 1. Unit of observation — patient, visit, service, case, dose, facility?
- 2. Definition — exactly what qualifies? Has it changed?
- 3. Grain — what is one row / one tick / one register entry?
- 4. Aggregation — what is summed, deduplicated or transformed?
- 5. Denominator — who should have received the service?
- 6. Missingness — absent because zero, not recorded, not reported or not applicable?
- 7. Incentives — what behaviour does the indicator or target encourage?
- 8. Provenance — can the published value be traced back to source documents?

Interactive 7 — Build a provenance specification

Groups of 3 • 6 minutes

- Choose one indicator from PHC: immunisation, ANC, TB, chronic care or another service.
- Write a one-page specification containing: definition, numerator, denominator, source document, aggregation rule, reporting period, geography and quality checks.
- Add one sentence: “This indicator could be biased if...”
- Then swap specifications with another group and try to break it.

From data to action



- The critical systems insight: action feeds back into the system and can change future data.
- Example: a dashboard identifies low coverage → managers intervene → service use and recording practices change → the next dataset reflects both the intervention and the measurement system.

Final challenge — Is the number real?

10-minute capstone discussion

- National dashboard: “TB treatment completion improved from 72% to 88%.”
- You are not allowed to accept or reject the claim yet.
- Ask: What is the definition? Numerator? Denominator? Which cohort? Which facilities? Completeness? Duplicates? Late reporting? Changes in recording? Targets/incentives? Data revision?
- Deliverable: a provenance chain and 3 tests that would increase confidence in the indicator.

Take-home message

For data scientists, DHIS data are generated — not simply collected

MEASUREMENT

Definitions determine what becomes visible.

TRANSFORMATION

Aggregation and denominators shape the indicator.

BEHAVIOUR

Targets, feedback and use can change future data.

- Best practice: never analyse a DHIS indicator without understanding its provenance.
- Ask not only “What does the number say?” but also “How was the number made?”

Link back to the original HIS lecture

- Health information systems support decision-making through data generation, compilation, analysis, communication and use.
- DHIS uses multiple collection tools, essential datasets, indicators, targets, data-quality checks, presentation, interpretation and feedback.
- The information cycle and decentralised use are central themes.
- This redesigned lecture adds a data-science lens: provenance, transformations, hidden assumptions, incentives and auditability.

Reference / source

- Rohde JE, Shaw V, Hedberg C, Stoops N, Venter S, Venter K, Matshisi L. Information for Primary Health Care. South African Health Review 2008, Chapter 13.